

Section I – Choose the correct answer

1[1] Computational Innovation Answer: A. A new or improved technology created using computing. ➡

Example: AI apps for healthcare.

2[2] Artificial Intelligence (AI) Answer: A. Simulating human intelligence in machines. ➡ *Example: ChatGPT, Siri.*

3[3] Machine Learning is a subset of: Answer: C. Artificial Intelligence (AI). ➡ *ML is how AI learns patterns from data.*

4[4] Defining characteristic of Big Data Answer: A. Its extremely large size and potential for revealing insights. ➡ *Example: analyzing millions of customer purchases.*

5[5] Telemedicine technology Answer: A. Telecommunications technology. ➡ *Video calls between doctors and patients.*

6[6] Main focus of EdTech Answer: C. Using technology to improve teaching and learning. ➡ *Example: Learning apps like Duolingo.*

7[7] Virtual Reality (VR) Answer: A. A simulated experience that can mimic or differ from reality. ➡ *Example: VR games.*

8[8] Computing tools for sustainability Answer: C. Renewable Energy Technology. ➡ *Example: AI optimizing solar power use.*

9[9] Example of Computational Innovation Answer: B. Software that automatically categorizes customer reviews using machine learning. ➡ *AI that sorts reviews into positive or negative.*

10[10] Tech enabling algorithms to improve automatically Answer: D. Machine Learning. ➡ *Example: Netflix recommendations.*

11[1] Computational Artifact Answer: C. A creation resulting from computing. ➡ *Example: Website, app, digital art.*

12[2] Stakeholder Answer: B. Any individual or group affected by a technology. ➡ *Users, developers, government.*

13[3] Ethical Impact Answer: D. The moral implications of using or deploying technology. ➡ *Example: AI bias, privacy issues.*

14[4] Bias in technology Answer: A. Systematic and unfair favoritism or prejudice. ➡ *AI hiring tools favoring one gender.*

15[5] Digital Divide Answer: C. The gap between those with and without access to technology. ➡ *Rural areas with no internet.*

16[6] Privacy Answer: D. The right to control personal information. ➡ *Controlling who can access your data.*

17[7] Data Collection Answer: C. The process of gathering and measuring information. ➡ *Websites collecting user behavior.*

1 **8** **Surveillance concerns Answer:** B. Privacy. ➡ *Example: CCTV monitoring.*

1 **9** **Misinformation Answer:** C. False or misleading information spread via digital platforms. ➡ *Fake news on social media.*

2 **0** **Accessibility in technology Answer:** C. The design of technology to be usable by people with diverse abilities. ➡ *Screen readers for visually impaired users.*

Section II – Written Question

Q: In what ways might the ongoing advancements in Artificial Intelligence exacerbate the existing digital divide, and what measures could be taken to mitigate these potential negative impacts?

Answer: AI advancements could widen the digital divide because people in underserved communities may lack access to AI tools, training, or infrastructure (internet, devices). This can lead to fewer opportunities and increase inequality. ➡ *For example: if AI-powered education apps are only available to wealthy schools, students in poorer areas fall further behind.*

Mitigation measures: ☒ Provide affordable internet and devices. ☒ Offer free or low-cost digital literacy and AI education programs. ☒ Develop AI tools that work on low-cost hardware. ☒ Encourage government policies that promote tech access for all.